① Find the binomial distribution which has mean 2 and variance $\frac{4}{3}$.

▲ We have $\mu = np$ and variance $= npq$

Given that $\mu = 2$ and variance $= \frac{4}{3}$

$\therefore np = 2$ and $npq = \frac{4}{3}$

$\Rightarrow (2)\, q = \frac{4}{3} \Rightarrow q = \frac{2}{3}$

$\therefore p = 1 - q = 1 - \frac{2}{3} = \frac{1}{3}$

Since $np = 2 \Rightarrow n\frac{1}{3} = 2 \Rightarrow n = 6$.

$\therefore P(x) = {}^nC_x\, p^x q^{n-x} = {}^6C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{6-x}$

=

② Let X be a binomially distributed random variable with mean 2 and S.D $\frac{2}{\sqrt{3}}$. Find the corresponding p.m.f.

▲ Given that $\mu = 2$; $S.D = \frac{2}{\sqrt{3}}$

We have $\mu = np$ and $S.D = \sqrt{npq}$

$\Rightarrow np = 2$ and $\sqrt{npq} = \frac{2}{\sqrt{3}} \Rightarrow npq = \frac{4}{3}$

$2q = \frac{4}{3}$

$q = \frac{2}{3}$

$\therefore p = \frac{1}{3}$

Since $np = 2 \Rightarrow n\left(\frac{1}{3}\right) = 2 \Rightarrow \boxed{n = 6}$

$\therefore$ p.m.f $P(x) = {}^6C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{6-x}$.

③ The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively. Find $P(x \geq 1)$.

▲ Given that $np = 4$ and $npq = \frac{4}{3}$

$4q = \frac{4}{3} \Rightarrow q = \frac{1}{3}$

$\therefore p = 1 - q = 1 - \frac{1}{3} = \frac{2}{3}$

$np = 4 \Rightarrow n\left(\frac{2}{3}\right) = 4 \Rightarrow \boxed{n = 6}$

$P(x) = {}^6C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{6-x}$

$P[x \geq 1] = 1 - P[x < 1] = 1 - P[x = 0] = 1 - {}^6C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^{6-0}$

$= 0.9986$.

④. Find the Probability of getting 4 heads in 6 tosses in a coin.

▲ Probability of getting a head in a single trail $= \frac{1}{2}$

$\therefore p = \frac{1}{2}$; $q = 1-p = \frac{1}{2}$

$n = 6$; $x = 4$

$$P(x) = {}^6C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{6-x}$$

$$P(4) = {}^6C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{6-4} = 15 \left(\frac{1}{2}\right)^6 = \frac{15}{64} = 0.23.$$

⑤. Determine the ~~bi~~ binomial distribution for which mean = 2 (Variance) and mean + variance = 3. Also find $P(x \leq 3)$.

▲ Given that $\mu = 2\sigma^2$

$$\Rightarrow np = 2npq \Rightarrow q = \frac{1}{2} \Rightarrow p = \frac{1}{2}$$

Also, we have: $\mu + \sigma^2 = 3$

$$\Rightarrow np + npq = 3 \Rightarrow np[1+q] = 3$$

$$n\left(\frac{1}{2}\right)\left[1+\frac{1}{2}\right] = 3 \Rightarrow n\left(\frac{1}{2}\right)\left(\frac{3}{2}\right) = 3$$

$\therefore \boxed{n = 4}$

$$P(x) = {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}$$

$$P[x \leq 3] = 1 - P[x > 3]$$

$$= 1 - P[x = 4]$$

$$= 1 - {}^4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{4-4}$$

$$= \frac{15}{16} = 0.9375.$$

⑥. Let X be a binomially distributed random variable based on 6 Bernoulli trails. If $p = 0.3$, Evaluate the the Probabilities $P(x=3)$, $P(x \leq 3)$, $P(x=4)$ and $P[x > 4]$

A Given that $n=6$ and $p=0.3$

$\therefore q = 1-p = 0.7$

$$P(x) = {}^{6}C_x [0.3]^x [0.7]^{6-x}$$

$$P[x=3] = {}^{6}C_3 [0.3]^3 [0.7]^{6-3} = 20 [0.3]^3 [0.7]^3 = 0.18522$$

$$P[x \le 3] = 1 - P[x>3] = 1 - \{P(x=4) + P(x=5) + P(x=6)\}$$

$$= 0.9295$$

$$P(x \ge 4) = {}^{6}C_4 [0.3]^4 [0.7]^{6-4} = 0.0595$$

$$P(x>4) = P[x=5] + P[x=6] = 0.0109.$$

(7) An ordinary six faced die is thrown four times. What are the probabilities of getting obtaining 4, 3, 2, 1 and 0 faces?

A A die is thrown 4 times.

$\therefore \boxed{n=4}$

A die has '6' faces.

$\therefore$ The probability of obtaining 'x' faces is

$$P(x) = {}^{4}C_x p^x q^{n-x}$$

Here $p = \frac{1}{6}$; $q = \frac{5}{6}$

$$\Rightarrow P(x) = {}^{4}C_x \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{4-x}$$

$$P(4) = {}^{4}C_4 \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^{4-4} = \left(\frac{1}{6}\right)^4$$

$$P(3) = {}^{4}C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{4-3} = 0.0154$$

$$P(2) = {}^{4}C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^{4-2} = 0.1157$$

$$P(1) = {}^{4}C_1 \left(\frac{1}{6}\right) \left(\frac{5}{6}\right)^{4-3} =$$

$$P(0) = {}^{4}C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^4 =$$

(8). If the Probability that a new born child is a male is 0.6, find the Probability that in a family of 5 children (i) there are exactly 3 boys (ii) at least two boys.

Given that $n = 5$ and $p = 0.6$

$\Rightarrow q = 0.4$ let x be the male child born.

$\therefore P(x) = {}^5C_x [0.6]^x [0.4]^{5-x}$

(i) $P[x=3] = {}^5C_3 [0.6]^3 [0.4]^{5-3} = 0.3456$

(ii) $P[x \geq 2] = 1 - P[x < 2] = 1 - \{P(x=0) + P(x=1)\}$

$= 0.913.$

(9). If on an average 1 vessel in every 10 is damaged, find the Probability that out of 5 vessels expected to arrive, at least 4 will arrive safely.

Given that $p = \frac{9}{10}$; $q = \frac{1}{10}$ and $n = 5$.

x : No. of vessels arrive safely.

$\therefore P(x) = {}^5C_x \left(\frac{9}{10}\right)^x \left(\frac{1}{10}\right)^{5-x}$

$P[x \geq 4] = P[x=4] + P[x=5]$

$= 0.91849.$

(10) The Probability that a bomb dropped from a plane will strike the target is $\frac{1}{5}$. If six bombs are dropped, find the Probability that (i) exactly two will strike the target (ii) at least two will strike the target.

Here $p = P(\text{hitting the target}) = \frac{1}{5}$

$\Rightarrow q = 1 - p = \frac{4}{5}$ and $n = 6$.

$P(x) = {}^6C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{6-x}$

(i) $P[x=2] = {}^6C_2 \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^{6-2}$

$= 0.246$

(ii) $P[x \geq 2] = 1 - P[x < 2]$

$= 1 - \{P[x=0] + P[x=1]\}$

$= 0.345$

(11). The probability that an entering student will graduate is 0.4. Determine the probability that out of 5 students (a) none (b) one and (c) at least one will graduate.

▲ Given that $p = 0.4 \Rightarrow q = 0.6$ and $n = 5$.

x : no. of students will be graduated

$P(x) = {}^5C_x [0.4]^x [0.6]^{5-x}$

(a) $P[x=0] = {}^5C_0 [0.4]^0 [0.6]^{5-0} = 0.07776$

(b) $P[x=1] = {}^5C_1 [0.4] [0.6]^{5-1} = 0.2592$

(c) $P[x \geq 1] = 1 - [P\, x < 1] = 1 - P[x=0] = 0.92224.$

(12). The no. of telephone lines busy at an instant time is a binomial variate with probability 0.2. If at an instant 10 lines are chosen at random, what is the probability that (i) 5 lines are busy (ii) at most 2 lines are busy (iii) All lines are busy?

▲ Given that $p = 0.2$; $q = 0.8$ and $n = 10$

x : no. of telephone lines busy at an instant

$\therefore P(x) = {}^{10}C_x [0.2]^x [0.8]^{10-x}$

(i) $P[x=5] = {}^{10}C_5 [0.2]^5 [0.8]^{10-5} = 0.0264$

(ii) $P[x \leq 2] = P[x=0] + P[x=1] + P[x=2] = 0.6776$

(iii) $P[x=10] = {}^{10}C_{10} [0.2]^{10} [0.8]^{10-10} = [0.2]^{10}.$

(13). The probability that a pen manufactured by a company will be defective is 0.1. If 12 such pens are selected, find the probability that (i) exactly 2 will be defective (ii) at least two will be defective (iii) None will be defective.

▲ $p = 0.1$; $q = 0.9$ and $n = 12$

x : no. of pens manufactured are defective

$P(x) = {}^{12}C_x (0.1)^x (0.9)^{12-x}$

(i) $P[x=2] = 0.2301$ (ii) $P[x \geq 2] = 1 - P[x < 2] = 0.3410$

(iii) $P(0) = 0.2824.$

(14). When a honest coin is tossed 4 times, find the probability of getting (i) exactly one head (ii) at most 3 heads (iii) at least two heads.

Given that $p = 0.5$; $q = 0.5$ and $n = 4$.

$P(x) = {}^4C_x (0.5)^x (0.5)^{4-x}$

x: no. of times getting head.

(i) $P[x=1] = {}^4C_1 [0.5]^1 [0.5]^{4-1} = 0.25$

(ii) $P[x \leq 3] = 1 - P[x > 3] = 1 - P[x=4] = 1 - {}^4C_4 (0.5)^4 [0.5]^{4-4}$

(iii) $P[x \geq 2] = 1 - P[x < 2] = 1 - \{P(x=0) + P(x=1)\}$

(15). In 256 sets of 12 tosses of a coin, in how many cases one can expect 8 heads and 4 tails.

$P[\text{head}] = \frac{1}{2} = p \Rightarrow q = 1 - p = \frac{1}{2} = P[\text{tail}]$

Given that $n = 12$

$\therefore P(x) = {}^{12}C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{12-x}$

By Binomial distribution, probability of 8 heads and 4 tails in 12 trails is

$P[x=8] = {}^{12}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^4 = \frac{495}{4096} = 0.1208$

$\therefore$ The expected no. of such cases in 256 sets

$= 256 \times P[x=8] = 256 \times 0.1208$

$= 30.9 \approx 31$

(a)

① It is known that 5% of the blades manufactured by a company are defective. The blades are supplied in packets of 10. Using the Binomial distribution find the probability that a packet choosen at random will contain
(i) at least 3 defective blades
(ii) at most 2 defective blades.

▶ Given that $p = \frac{5}{100} = 0.05$ $\therefore q = 1-p = 1-0.05 = 0.95$ and $n = 10$.

Let x denote the defective blades.

$\therefore P(x) = {}^{n}C_x\, p^x\, q^{n-x}$

$$\boxed{P(x) = {}^{10}C_x\, [0.05]^x\, [0.95]^{10-x}}$$

(i) $P[\text{at least 3 defective blades}] = P[x \geq 3]$

$= 1 - P[x < 3]$

$= 1 - \{P(x=0) + P(x=1) + P(x=2)\}$

$= 1 - \{{}^{10}C_0\,[0.05]^0\,[0.95]^{10-0} + {}^{10}C_1\,[0.05]^1\,[0.95]^9 + {}^{10}C_2\,[0.05]^2\,[0.95]^8\}$

$= 0.011$

(ii) $P[\text{at most 2 defective blades}]$

$= P[x \leq 2] = P[x=0] + P[x=1] + P[x=2]$

$= {}^{10}C_0\,[0.05]^0\,[0.95]^{10} + {}^{10}C_1\,[0.05]^1\,[0.95]^9 + {}^{10}C_2\,[0.05]^2\,[0.95]^8$

$= 0.987.$

② A box containing 100 transistions, 20 of which are defective and 10 are selected for inspection. Calculate the probability that (i) at most one defective (ii) at least 4 defective.

▶ Given that $p = \frac{20}{100} = \frac{1}{5} = 0.2$; $q = 1-p = 1-0.2 = 0.8$

and $n = 10$. Let x denote the defective transtions.

$\therefore P(x) = {}^{10}C_x\,[0.2]^x\,[0.8]^{10-x}$

(i) $P[\text{at most one defective}] = P[x \leq 1] = P[x=0] + P[x=1]$

$= {}^{10}C_0\,[0.2]^0\,[0.8]^{10} + {}^{10}C_1\,[0.2]^1\,[0.8]^9 = 0.375.$

(ii). $P[\text{at least 4 defective}]$

$= P[x \geq 4]$

$= 1 - P[x < 4]$

$= 1 - \{P[x=0] + P[x=1] + P[x=2] + P[x=3]\}$

$= 1 - \{10c_0 [0.2]^0 [0.8]^{10} + 10c_1 [0.2]^1 [0.8]^9 + 10c_2 [0.2]^2 [0.8]^8 + 10c_3 (0.2)^3 [0.8]^7\}$

$= 0.122$.

③. The Probability that a man aged 60 will live to be 70 is 0.65. What is the Probability that out of 10 men, now aged 60, (i) exactly 9 will live to be 70 (ii) at most 9 will live to be 70 and (iii) at least 7 will live to be 70?

▲ Given that $p = 0.65$; $q = 1-p = 1-0.65 = 0.35$ and $n=10$

Let x denote the men will live to be 70.

$\therefore P(x) = {}^nC_x\, p^x q^{n-x}$

$P(x) = 10c_x (0.65)^x (0.35)^{10-x}$

(i) $P[\text{exactly 9 will live to be 70}] = P[x=9] = 10c_9 [0.65]^9 [0.35]^{10-9} = 0.0725$

(ii) $P[\text{at most 9 will live to be 70}]$

$= P[x \leq 9] = 1 - P[x>9] = 1 - \{P(x=10)\}$

$= 1 - 10c_{10} [0.65]^{10} [0.35]^{10-10} = 1 - 0.0135 = 0.9865$.

(iii) $P[\text{at least 7 will live to be 70}]$

$= P[x \geq 7] = P[x=7] + P[x=8] + P[x=9] + P[x=10]$

$= 10c_7 [0.65]^7 (0.35)^3 + 10c_8 (0.65)^8 (0.35)^2 + 10c_9 (0.65)^9 (0.35) + 10c_{10} (0.65)^{10} (0.35)^{10-10}$

$= 0.2522 + 0.1757 + 0.0725 + 0.0135 = 0.5139$.

④. In a quiz contest of answering 'Yes' or 'No', what is the Probability of guessing at least 6 answers correctly out of 10 questions asked? Also find the probability of the same if there are 4 options for a correct answer.

▲ (i) Let x denote the correct answer and we have in the first case $p = \frac{1}{2}$; $q = \frac{1}{2}$ and $n = 10$

$\therefore P(x) = 10c_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x}$

Probability of guessing at least 6 answers correctly out of 10 questions is $= P[x \geq 6] = P[x=6] + P[x=7] + P[x=8] + P[x=9] + P[x=10]$

$= 10c_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^4 + 10c_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3 + 10c_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^2 + 10c_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right) + 10c_{10} \left(\frac{1}{2}\right)^{10}$

$= 0.377$.

(ii) In the second case when there are 4 options:

$$p = \frac{1}{4} \; ; \quad q = 1-p = \frac{3}{4} \quad \text{and} \quad n = 10$$

$$\therefore P(x) = {}^{10}C_x \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{10-x}$$

$$P[x \geq 6] = P[x=6] + P[x=7] + P[x=8] + P[x=9] + P[x=10]$$

$$= {}^{10}C_6 \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^4 + {}^{10}C_7 \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^3 + {}^{10}C_8 \left(\frac{1}{4}\right)^8 \left(\frac{3}{4}\right)^2 + {}^{10}C_9 \left(\frac{1}{4}\right)^9 \left(\frac{3}{4}\right)^1$$

$$+ {}^{10}C_{10} \left(\frac{1}{4}\right)^{10} = 0.019.$$

⑤. In sampling a large number of parts manufactured by a machine, the mean number of defectives in a sample of 20 is 2. Out of 1000 such samples, how many would be expected to contain at least 3 defective parts.

▲ Mean number of defectives $= np = 2$

Given that $n = 20$. This $\Rightarrow np = 2$

$$p = \frac{2}{n} = \frac{2}{20} = \frac{1}{10} = 0.1$$

$\Rightarrow q = 1-p = 1-0.1 = 0.9$

Let x denote the defective parts.

$$P(x) = {}^{20}C_x (0.1)^x (0.9)^{20-x}$$

$\therefore$ The probability of at least three defectives in a sample of 20 = $P[x \geq 3]$

$$= 1 - P[x < 3]$$

$$= 1 - \{P(x=0) + P(x=1) + P(x=2)\}$$

$$= 1 - \{{}^{20}C_0 (0.1)^0 (0.9)^{20} + {}^{20}C_1 (0.1)^1 (0.9)^{19} + {}^{20}C_2 (0.1)^2 (0.9)^{18}\}$$

$$= 0.323$$

Thus, the no. of samples having at least three defective parts out of 1000 samples $= 1000 \times 0.323 = 323$.

⑧. Out of 800 families with 5 children each, how many would you expect to have (a) 3 boys (b) 5 girls (c) either 2 or 3 boys? Assume equal probabilities for boys and girls.

▶ Given that $n = 5$; $p = 0.5$; $q = 0.5$

Let x denote the boys children.

$$P(x) = {}^{5}C_{x} (0.5)^{x} (0.5)^{5-x}$$

(i) $P[x=3] = {}^{5}C_{3} (0.5)^{3} (0.5)^{5-2} = 0.3125$

out of 800 families it $= 800 \times 0.3125 = 250$

(ii) $P[\text{5 girls}] = P[x=0] = {}^{5}C_{0} (0.5)^{0} (0.5)^{5-0} = 0.03125$

out of 800 families it $= 800 \times 0.03125 = 25$

(iii) $P[\text{2 or 3 boys}] = P[x=2] + P[x=3]$

$= {}^{5}C_{2} (0.5)^{2} (0.5)^{5-2} + {}^{5}C_{3} (0.5)^{3} (0.5)^{5-3} = 0.625$

out of 800 family it $= 800 \times 0.625 = 500$

(10). Two persons A and B play a game in which their chances of winning are the ratio 3:2. If 6 games are played, find A's chances of winning at least three games.

Given that $n = 6$; $p = \frac{3}{5}$; $q = \frac{2}{5}$

Let x denote the success of 'A'.

$$P(x) = {}^6C_x \left(\frac{3}{5}\right)^x \left(\frac{2}{5}\right)^{6-x}$$

$$P[x \geq 2] = 1 - P[x < 2]$$

$$= 1 - \{P[x=0] + P[x=1]\}$$

$$= 1 - \left\{ {}^6C_0 \left(\frac{3}{5}\right)^0 \left(\frac{2}{5}\right)^6 + {}^6C_1 \left(\frac{3}{5}\right)^1 \left(\frac{2}{5}\right)^5 \right\}$$

$$= 0.8206$$

(11). An airline knows that 5% of the people making reservation on a certain flight will not turn up. Consequently, their policy is to sell 52 tickets for a flight that can hold 50 passengers. What is the probability that there will be a seat available for every passenger who turns up?

A Given that $p = \frac{5}{100} = 0.05$: $q = 0.95$ and $n = 52$

x : no. of passengers will not turn up.

$$P(x) = {}^{52}C_x (0.05)^x (0.95)^{52-x}$$

$$P[x \geq 2] = 1 - P[x < 2] = 1 - \{P[x=0] + P[x=1]\}$$

$$= 1 - \{ {}^{52}C_0 (0.05)^0 (0.95)^{52} + {}^{52}C_1 (0.05)^1 (0.95)^{51} \}$$

$$= 0.7405$$

(12). In a consignment of electric lamp, 5% are defective. A random sample of 8 lamps is taken for inspection. What is the probability that it has one or more defectives?

A Given that $n = 8$: $p = \frac{5}{100} = 0.05 \Rightarrow q = 1 - p = 0.95$

Let x denote the defective lamps.

$$\therefore P(x) = {}^{8}C_x (0.05)^x (0.95)^{8-x}$$

$$P[x \geq 1] = 1 - P[x < 1] = 1 - P[x=0] = 1 - {}^{8}C_0 (0.05)^0 (0.95)^8 = 0.3365.$$